

Shahan Ayyubi

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Education

Jamia Millia Islamia, New Delhi

2024 – 2028

Bachelor of Technology in Computer Engineering (CGPA: 8.81/10)

Skills

Languages: C++, Python, JavaScript, TypeScript, SQL

Frameworks: React, Node.js, Express, Django, REST APIs, Tailwind CSS

Data & ML: PyTorch, NumPy, Pandas, Scikit-learn, CNN, LSTM

Tools & Platforms: Git, GitHub, GCP, Linux, MongoDB, Vercel, Railway

Projects

Sparrow | *React, Django, Leaflet, Turf.js* | [Repository](#) | [Live](#)

Mar 2026

- Engineered a ward-level AQI dashboard for Delhi, rendering real-time data across 250+ municipal wards on choropleth maps with GeoJSON polygon boundaries.
- Designed a Django REST API with geospatial point-in-polygon queries (Turf.js) to map sensor readings to ward polygons, enabling sub-city granularity for analysis.
- Integrated Recharts for historical AQI trend visualization and GSAP scroll-driven animations; deployed frontend on Vercel and backend on Railway.

Project Canary | *PyTorch, CNN, BiLSTM, Attention* | [Repository](#)

Aug – Oct 2025

- Architected a deep learning pipeline fusing Sentinel-5P satellite imagery, ERA5 meteorological data, and CPCB ground-station readings to forecast ground-level NO₂ and O₃ concentrations.
- Engineered a CNN-BiLSTM architecture with temporal attention over 24-hour sliding windows, processing spatial raster inputs of shape $(T, 9, 10, 11)$ alongside 18 tabular weather features.
- Achieved R² of 0.76 (NO₂) and 0.67 (O₃) on held-out test data, outperforming XG-Boost baseline by ~10% in RMSE.

Cassetto | *Python, tree-sitter, DuckDB, LanceDB* | [Repository](#) | [PyPI](#)

May 2026

- Built a local code intelligence engine that parses 13 languages via tree-sitter ASTs into a call graph (DuckDB), vector store (LanceDB), and keyword index (SQLite FTS5), exposing 18 tools to AI assistants over MCP.
- Implemented hybrid retrieval (BM25 + semantic vectors) with Reciprocal Rank Fusion and graph-aware reranking, achieving 92% precision and 91% recall.
- Engineered graph analysis tools with sub-30ms latency; published to PyPI with cross-platform CI/CD across 3 OSes and 3 Python versions.

Experience

Research Intern, *Indraprastha Institute Of Information Technology, Delhi*

Jun 2026 – Present

- Conducting research on transformer optimization, designing and evaluating techniques to improve inference efficiency, memory utilization, and model throughput.

Cloud Lead, *Google Developer Groups on Campus, JMI*

Aug 2025 – May 2026

- Led a 5-member team to organize 2 GC workshops (75+ attendees), driving record Study Jam participation of 150+ students and a 2× YoY increase across the chapter.

Founder, *teamV (Creative Agency)*

Jul 2024 – Jan 2026

- Managed a 5-person freelance team, delivering 30+ marketing projects for 10+ clients, generating INR 3L+ in revenue while owning the end-to-end project lifecycle.

Achievements

LeetCode: 550+ problems solved (top 15% globally)

HackFinance, IIITD: Qualified for the final round out of 100+ teams

Certifications: CS50x (Harvard), Google Cloud Study Jam, DevOps Essentials (IBM)